

Introduction to R and RStudio

Part 1: Basics (Follow Along in RStudio)

Dr. Jomana Bashatah

Learning Outcomes

- What you will learn in this session:
- How to install R and RStudio
- What is the windows layout of RStudio
- How to setup RStudio
- How to create a project (folder) in RStudio
- How to use major functionalities of RStudio
- How to extend R's functionality with R-packages
- Which packages you should install for this course
- Data types and data objects in R
- How very big and very small numbers can be displayed

Install and Setup R and RStudio

- A typical setup to work with R consists of two components:
- The R Console which executes R code
- An integrated development environment (IDE) such as RStudio
- You can download and install R and RStudio by following the steps in:
- <https://ise423-25.blogspot.com/2025/09/installing-r-and-rstudio-complete-step.html>

RStudio - Integrated Development Environment (IDE)

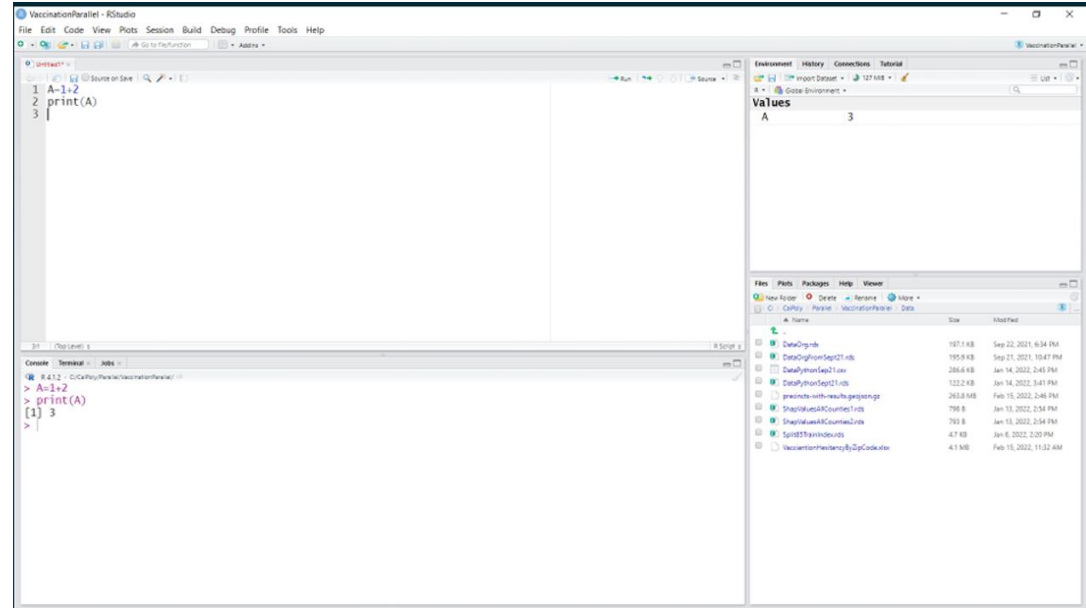
RStudio provides a user-friendly interface with four main panes:

Source Editor (top-left): Write and edit R scripts

Console (bottom-left): Execute R commands directly

Environment/History (top-right): View variables and command history

Files/Plots/Packages/Help (bottom-right): Manage files, view plots, install packages



RStudio Window

Recommended RStudio Settings

- Essential Setup Steps:
- Do not Restore .RData into workspace at startup:
 - Go to Tools → Global Options
 - Uncheck “Restore .RData into workspace at startup”
- Work with R Projects:
 - This assigns a directory on your hard drive to your R analysis
 - File → New Project

R Packages

- R Packages extend R's functionality. They have to be installed only once:
- # Install packages using Tools -> Install Packages... or:
`install.packages(c("tidyverse", "rio"))`
- After installation they need to be loaded every new R script:
- `library(rio)`
`library(tidyverse)`
- Please install these packages soon:
- tidyverse: supports easy data processing
- rio: allows loading various data resources with one `import()` command
- janitor: provides functionality to clean data and rename variable names
- tidymodels: streamlines data engineering and machine learning tasks
- kableExtra: supports rendering tables in HTML
- shiny: needed together with the learnr package for interactive exercises

Example: The rio and tidyverse Package

- `library(rio); library(tidyverse)`

```
DataHousing = import("Data/HousingData.csv") |>  
  select(Price=price, Sqft=sqft_living,  
         Bedrooms=bedrooms, Waterfront=waterfront)
```

```
print(DataHousing[1:3,])
```

- | | Price | Sqft | Bedrooms | Waterfront |
|---|--------|------|----------|------------|
| 1 | 221900 | 1180 | 3 | no |
| 2 | 538000 | 2570 | 3 | no |
| 3 | 180000 | 770 | 2 | no |
- Note: `import()` requires rio package; `select()` requires tidyverse package

Data Types & Data Objects Overview

- Data Types: What can R store?
- numerical (num)
- character (chr)
- factor
- logic
- Data Objects: What containers does R use?
- single entry - single variable
- list of entries - vectors
- table - dataframe and tibble
- advanced objects - plots, models, prediction results

Data Types

- Numerical Data Type (num): Numerical values (e.g., 1, 523, 3.45) used for calculation. ZIP-Codes are NOT numerical data type.
- Character Data Type (chr): Storing sequence of characters, numbers, and/or symbols (e.g. first/last names, street addresses, Zip-codes)
- Factor Data Type (factor): A factor is an R data type that stores categorical data in an effective way. factor data types are also required by many classification models in R.
- Logic Data Type (logic): A data type that stores the logic states TRUE and FALSE is called a logic object (sometimes called Boolean)

Data Types - Numerical

- Numerical values are used for calculations (therefore ZIP-Codes are not numerical). Numerical data can be discrete (integer) or continuous (double).
- `A = as.integer(2)`
`B = as.integer(3)`
`str(A)` # `str()` returns structure of a variable
- `int 2`
- `C = 1.23`
`str(C)`
- `num 1.23`
- `print(A * C)`
- `[1] 2.46`
- `A^B`
- `[1] 8`
- `A/B` # Returns num type
- `[1] 0.6666667`

Data Types - Character

- Note: What is called a “character” in R is often called a “string” in other programming languages.
- Character data types must be surrounded by quotes:
- `MyText = "Hello world!"`
`print(MyText)`
- `[1] "Hello world!"`
- Character variables can be concatenated with the `cat()` command:
- `FirstName = "Jomana"`
`LastName = "Bashatah"`
`cat(FirstName, LastName) # R adds a space automatically`
- Jomana Bashatah

Data Types - Factor

- A factor stores categorical data efficiently. Categorical data are character type data covering a few categories such as hair color (blonde, brown, red, black). They can be coded with numbers (e.g., 1-5 for hair color) and use less memory.
- ```
People=tribble(~Name,~Sex,
 "John", "male",
 "Jane", "female",
 "Mia", "female",
 "Brid", "female",
 "Greg", "male")
print(People)
```
- # A tibble: 5 × 2
- Name Sex
- <chr> <chr>
- 1 John male
- 2 Jane female
- 3 Mia female
- 4 Brid female
- 5 Greg male

# Data Types - Logic

- Logic variables store TRUE and FALSE. They can be combined with and/or. Internally TRUE is stored as 1 and FALSE is stored as 0.
- `IsConcertGood = FALSE`  
`IsCompanyGood = TRUE`  
`cat("Is the concert good?", IsConcertGood, "Is the company good?", IsCompanyGood)`
- Is the concert good? FALSE Is the company good? TRUE
- `IsEveningAmazing=IsConcertGood & IsCompanyGood`  
`cat("Is the evening amazing?", IsEveningAmazing)`
- Is the evening amazing? FALSE
- `IsEveningGood = IsConcertGood | IsCompanyGood # | stands for "or"`  
`cat("Is the evening good?", IsEveningGood)`
- Is the evening good? TRUE
- `IsConcertGood+IsCompanyGood+17`
- [1] 18

# Data Types – Truth Table

| R Object A | R Object B | A & B | A   B |
|------------|------------|-------|-------|
| TRUE       | TRUE       | TRUE  | TRUE  |
| TRUE       | FALSE      | FALSE | TRUE  |
| FALSE      | TRUE       | FALSE | TRUE  |
| FALSE      | FALSE      | FALSE | FALSE |

# Data Objects

- Single Value Object: Stores just one value
- Vector Object: Stores a list of values
- Data Frame (Tibble) Object: Table-like structure
- List Object: (not covered in this course)
- Advanced Objects: plots, models, recipes

# Single Value Object

- Object stores a single value:
- A = 123.768  
B = 3  
C = "Hello World"  
IsLifeGood = TRUE



# Vector Objects

- A vector object stores a list of values (numerical, character, factor, or logic).
- Example: Weather during the last three days in Stattown:  
VecTemp = c(70, 68, 55)  
VecWindSpeed = c("low", "low", "high")  
VecIsSunny = c(TRUE, TRUE, FALSE)
- Vector objects can be used as arguments for an R command to calculate
- MeanForecTemp=mean(VecTemp)  
cat("The average forecasted temperature is", MeanForecTemp)
- The average forecasted temperature is 64.33333
- ForecDays=length(VecTemp)  
cat("The forecast is for", ForecDays, "days.")
- The forecast is for 3 days.

## Data Frames (Tibbles)

A data frame is similar to an Excel table:

| Survived | Pclass | Sex    | Age | FareInPounds |
|----------|--------|--------|-----|--------------|
| 0        | 3      | male   | 22  | 7.2500       |
| 1        | 1      | female | 38  | 71.2833      |
| 1        | 3      | female | 26  | 7.9250       |
| 1        | 1      | female | 35  | 53.1000      |
| 0        | 3      | male   | 35  | 8.0500       |

# Extracting Vectors and Performing Calculations

- Numerical Vectors
- `library(rio); library(tidyverse)`  
`DataTitanic = import("Data/Titanic.csv")`
- `VecFareInPounds = DataTitanic$FareInPounds`  
`AvgFare = mean(VecFareInPounds)`  
`cat("The average fare of Titanic passengers was:", AvgFare,`  
`"British Pounds")`
- The average fare of Titanic passengers was: 32.30542 British Pounds

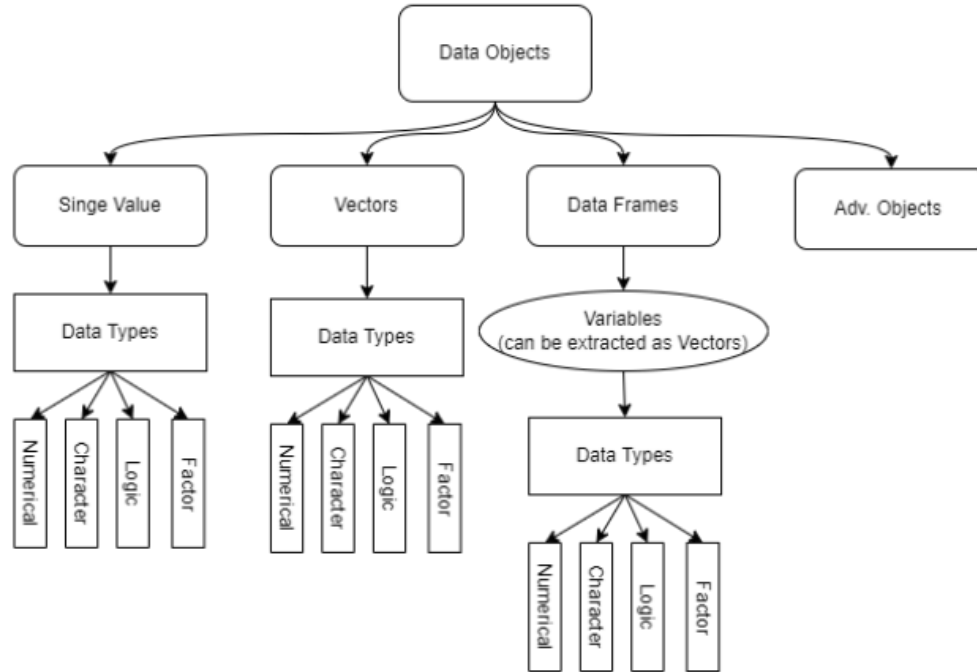
# Extracting Vectors and Performing Calculations

- Logical Vectors
- `DataTitanic$Survived = as.logical(DataTitanic$Survived)`  
`str(DataTitanic)`
- 'data.frame': 887 obs. of 8 variables:
- `$ Survived` : logi FALSE TRUE TRUE TRUE FALSE FALSE ...
- `$ Pclass` : int 3 1 3 1 3 3 1 3 3 2 ...
- `$ Name` : chr "Mr. Owen Harris Braund" "Mrs. John Bradley Cumings" "Miss. Laina Heikkinen" "Mrs. Jacques Heath Futrelle" ...
- `$ Sex` : chr "male" "female" "female" "female" ...
- `$ Age` : num 22 38 26 35 35 27 54 2 27 14 ...
- `$ SiblingsSpousesAboard`: int 1 1 0 1 0 0 0 3 0 1 ...
- `$ ParentsChildrenAboard`: int 0 0 0 0 0 0 0 1 2 0 ...
- `$ FareInPounds` : num 7.25 71.28 7.92 53.1 8.05 ...
- `SurvRate = mean(DataTitanic$Survived)`  
`cat("The average survival rate of Titanic passengers was:", SurvRate)`
- The average survival rate of Titanic passengers was: 0.3855693

# Data Frames vs. Tibbles

- A tibble is a more advanced sub-type of a data frame. Convert with `as_tibble()`:
- Key Differences:
- A few of the differences between data frames and tibbles:
- A data frame outputs all its rows and columns by default. A tibble outputs only the first 10 rows and the variables that fit on the screen but provides information about omitted variables and rows.
- A data frame can have row names, while a tibble cannot.
- In R version <4.1 a data frame converts all character values to factor type. This conversion was often confusing and annoying. In contrast, a tibble only coerces character values into factor on demand. Since R version 4.1 regular data frames behave the same as tibbles.

# Summary: Data Types and Objects



# How Are Very Big Numbers Presented

- The GDP for 2021 in the US was \$22,996,086,000,000 (rounded to millions)

$$GDP = 2.2996086 \cdot 10000000000000$$

$\Leftrightarrow$

- $GDP = 2.2996086 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10$

$\Leftrightarrow$

$$GDP = 2.2996086 \cdot 10^{13}$$

- What R does:
- `GDPUS = 22996086000000`  
`print(GDPUS)`
- `[1] 2.299609e+13`

# How Are Very Small Numbers Presented

- The probability of getting struck by lightning in the US is about 0.000000000365 on any randomly chosen day.

$$\begin{aligned}
 ProbLight &= \frac{3.65}{10000000000} \\
 &\Leftrightarrow \\
 ProbLight &= \frac{3.65}{10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10 \cdot 10} \\
 &\Leftrightarrow \\
 ProbLight &= \frac{3.65}{10^{10}} \\
 &\Leftrightarrow \\
 ProbLight &= 3.65 \cdot 10^{-10}
 \end{aligned}$$

Why?

- In the U.S., a person has a 1:10,000-lifetime risk of being struck by lightning. Assuming a life span of 75 years and 365.25 days per year, the probability per day is:

$$\begin{aligned}
 \frac{1}{10,000 \cdot 365.25 \cdot 75} &= \frac{1}{273,937,500} \\
 &= 0.000000000365 \\
 &= 3.65 \times 10^{-10}
 \end{aligned}$$

- What R Does
- `ProbStruck = 0.000000000365`  
`cat("Probability to get struck by lightning on an avg. day:", ProbStruck)`
- Probability to get struck by lightning on an avg. day: 3.65e-10